“Chapter # 1”

A programmer perspective: An Introduction to computer

Q #1: Difference between text file and binary file?

- Text files are used to store data more user friendly.
- ❖ Binary files are used to store data more compactly.
- In the text file, a special character whose ASCII value is 26 inserted after the last character to mark the end of file.
- ❖ In the binary file no such character is present. Files keep track of the end of the file from the number of characters present.
- Content written in text files is human readable.
- ❖ Content written in binary files is not human readable and looks like encrypted content.

Q #2: What is the difference between buffer and memory?

Buffer is an area of memory used to temporarily store data while it's being moved from one place to another. Cache is a temporary storage area used to store frequently accessed data for rapid access.

Q # 3: What is kernel with example?

It is the core that provides basic services for all other parts of the OS. It is the main layer between the OS and underlying computer hardware, and it helps with tasks such as process and memory management, file systems, device control and networking. A kernel is a part of the operating system that converts user commands into machine language.

In computing, the kernel is the central component of an operating system that manages system resources and provides low-level services to other software components. The kernel is responsible for managing hardware resources such as CPU, memory, disk I/O, network devices, and input/output devices, and for providing a set of system calls that can be used by applications to access these resources.

The kernel acts as an intermediary between the hardware and software components of a computer system.

Q # 4: What is meant by shell?

A shell is a computer program that presents a command line interface which allows you to control your computer using commands entered with a keyboard instead of controlling graphical user interfaces (GUIs) with a mouse/keyboard/touch screen combination.

Q # 5: What is the meaning of DMA?

Direct Memory Access

Direct Memory Access (DMA) is a capability provided by some computer bus architectures that allows data to be sent directly from an attached device (such as a disk drive) to the memory on the computer's motherboard.

Q # 6: What is meant by Buses?

A bus is a high-speed internal connection. Buses are used to send control signals and data between the processor and other components.

There are three types of bus lines: Data bus, Address bus, and Control bus.

Q # 7: What is SRAM (static random access memory)?

SRAM (static RAM) is a type of random access memory (RAM) that retains data bits in its memory as long as power is being supplied.

Q # 8: what is meant by Uniprocessor?

A Uniprocessor refers to a computer system that has a single central processing unit (CPU) that is responsible for executing instructions and performing calculations.

In other words, it has only one core or processing unit, which means it can only perform one task at a time.

This is in contrast to a multiprocessing system, which has multiple CPUs or cores that can work together to execute multiple tasks simultaneously.

Q # 9: What is meant by context switching?

Context switching refers to a technique/method used by the OS to switch processes from a given state to another one for the execution of its function using the CPUs present in the system.

Q # 10: What is context switching in OS ?

Context Switching involves storing the context or state of a process so that it can be reloaded when required and execution can be resumed from the same point as earlier. This is a feature of a multitasking operating system and allows a single CPU to be shared by multiple processes.

Q # 11: what is meant by multiprocessor?

A multiprocessor system refers to a computer architecture that has multiple central processing units (CPUs) or processors that work together to perform computational tasks.

These processors can share memory and other resources, allowing them to perform tasks in parallel and achieve higher performance than a uniprocessor system.

Q # 12: What is multithreading vs multiprocessing?

By formal definition, multithreading refers to the ability of a processor to execute multiple threads concurrently, where each thread runs a process. Whereas multiprocessing refers to the ability of a system to run multiple processors concurrently, where each processor can run one or more threads.

Q # 13: What is meant by thread?

A thread refers to a sequence of instructions that can be executed independently by a single process. In other words, a thread is a lightweight process that can run concurrently with other threads within the same process, sharing the same memory space and system resources.

Threads are commonly used in modern operating systems to enable multitasking and improve application performance. By creating multiple threads within a single process, an application can perform multiple tasks simultaneously, allowing for better resource utilization and faster execution.

Q # 14: what is meant by virtual machine?

A virtual machine (VM) refers to a software emulation of a physical computer system that can run its own operating system and applications. In other words, a VM creates a virtualized environment that behaves like a real computer, with its own virtualized CPU, memory, storage, and I/O devices.

The virtualization technology allows multiple VMs to run on a single physical machine, each with its own isolated and secure environment. This can improve resource utilization, reduce hardware costs, and simplify system administration.

Q # 15: What is meant by files in Unix?

In Unix, a file is a basic unit of storage that represents a collection of related data or information. A file can contain any type of data, such as text, binary, images, audio, or video.

Files are organized into a hierarchical file system, which is a tree-like structure that organizes files and directories (also known as folders) into a logical structure.

Q # 16: Difference between concurrency and parallelism?

Concurrency is about multiple tasks which start, run, and complete in overlapping time periods, in no specific order. Parallelism is about multiple tasks or subtasks of the same task that literally run at the same time on a hardware with multiple computing resources like multi-core processor.

Q # 17: What is meant by Multicore Processor?

Multicore refers to a computer processor that contains multiple processing units (cores) on a single chip. Each core is a complete central processing unit (CPU) capable of executing its own instructions independently of the other cores. A multicore processor can execute multiple instructions simultaneously, increasing the overall processing power of the computer.

Question: What is meant by Multithreading?

Multithreading is the ability of a program or an operating system to enable more than one user at a time without requiring multiple copies of the program running on the computer.

Multithreading can also handle multiple requests from the same user.

Q # 18: What is meant by Hyper_threading?

Hyper-threading is a process by which a CPU divides up its physical cores into virtual cores that are treated as if they are actually physical cores by the operating system. These virtual cores are also called threads . Most of Intel's CPUs with 2 cores use this process to create 4 threads or 4 virtual cores.

Q # 19: What is SIMD parallelism?

SIMD (Single Instruction, Multiple Data) parallelism is a type of parallel processing that allows a single instruction to operate on multiple pieces of data simultaneously. This technique is commonly used in multimedia processing, scientific simulations, and other applications that require large amounts of data to be processed in parallel.

In SIMD parallelism, a single instruction is applied to a vector of data elements, such as a set of pixels in an image, a sequence of audio samples, or a matrix of values in a mathematical operation. The advantage of SIMD parallelism is that it allows for high-speed processing of large amounts of data, without requiring complex parallel programming techniques. Examples of SIMD instructions include vector addition, multiplication, and logical operations, as well as specialized operations for image and signal processing.